

D468

Discovering Data

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

The operations lead at a 35-location car wash chain, which recently rebranded, asks: “How can we make customers happier?” An analyst in her second week wants to reframe it as a SMART question. Which is best?

- A. Which two changes to wait time or pricing would raise the post-visit rating from 3.8 to 4.2 at our 10 lowest-rated locations by June 30?
- B. What is our average customer rating?
- C. How do customers feel about us?
- D. Why do customers dislike our new brand?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which SMART element is missing from this question: “Which of our three marketing channels should we increase spending on, based on cost per first-time customer?”

- A. Specific
- B. Measurable
- C. None; it is fully SMART.
- D. Time-bound

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A grocery chain with 120 stores collected loyalty card data, removed duplicate and incomplete records, and fixed date formats. The data is stored in a cloud warehouse. What phase comes next?

- A. Analyze
- B. Act
- C. Share
- D. Prepare

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A hotel group with 18 properties and a new loyalty program gathers four fields per stay: guest comments on what they liked, number of nights, room type (standard or suite), and check-in wait in minutes. Which fields are qualitative?

- A. All four, since all come from guests.
- B. Guest comments only.
- C. Nights and wait minutes.
- D. Guest comments and room type.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An analyst wants to understand why customers cancel their subscriptions. Which data best answers “why”?

- A. Monthly cancellation counts.
- B. Only quantitative data, because qualitative data cannot be analyzed.
- C. Qualitative data such as exit interviews and open-ended comments.
- D. Structured billing logs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A survey to a regional bank’s 9,000 mobile app users, sent the week the app won a design award, asks: “How much do you love our new app?” What is wrong with the question?

- A. It should wait until the award publicity fades.
- B. It is leading; ask neutrally, such as “How satisfied are you with the app?”
- C. 9,000 users is too few to survey.
- D. It is closed-ended.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A food bank with 400 volunteers set the goal “reduce food waste by 15% this year.” Its director drives a delivery route every Friday. Which is the best KPI?

- A. Number of active volunteers.
- B. Pounds of food discarded as a share of pounds received, tracked monthly.
- C. Total pounds of food distributed.
- D. Hours the director spends on deliveries.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An insurer with 1.2 million policyholders keeps policy records in a relational database and receives thousands of claim photos and adjuster voice notes each week. Its claims unit moved to a new office in April. Which data are unstructured?

- A. None; any digital file is structured.
- B. The claim photos and voice notes.
- C. Only the voice notes.
- D. The policy records.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A dataset has a fixed schema of rows and columns and can be queried with SQL. Which term fits?

- A. Qualitative data
- B. Unstructured data
- C. Structured data
- D. Semi-structured data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A fulfillment center that runs three shifts and ships 25,000 orders a day sees late shipments rise. The team asks why: pickers are slow because items are in the wrong bins, because putaway staff skip scanning, because scanners die mid-shift, because no one is assigned to charge them. The site manager holds a lean certification. What root cause should be fixed?

- A. No one is assigned to charge the scanners.
- B. Pickers are slow.
- C. Items are in the wrong bins.
- D. Retrain putaway staff to scan.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A real estate analyst at a firm founded in 1995 lists five sale prices on one street: \$210,000, \$225,000, \$230,000, \$240,000, and \$1.2 million for a lakefront estate. The street has mature oak trees. Which measure best represents a typical price, and what is it?

- A.** Mean, \$421,000
- B.** Midrange, \$705,000
- C.** Median, \$230,000
- D.** \$1.2 million, the highest sale

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

Five salaries have a median of \$60,000. The highest salary rises from \$90,000 to \$300,000. What happens to the median and the mean?

- A.** The median rises; the mean stays the same.
- B.** The median stays \$60,000; the mean rises by \$210,000.
- C.** Both rise by \$42,000.
- D.** The median stays \$60,000; the mean rises by \$42,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

The operations lead at a 35-location car wash chain, ~~which recently rebranded,~~ asks: “How can we make customers happier?” An analyst ~~in her second week~~ wants to reframe it as a SMART question. Which is best?

Target word: best SMART question **Decides:** It must be specific, measurable, actionable, relevant, and time-bound.

A	Correct	Specific, measurable, action-oriented, relevant, and time-bound.
B	Adjacent category	Measurable, but it leads to no action and has no time frame.
C	Dropped element	Not measurable, actionable, or time-bound.
D	Camouflage	Chases the rebrand detail and assumes the answer.

2. Answer D Concept 1 • transfer

Which SMART element is missing from this question: “Which of our three marketing channels should we increase spending on, based on cost per first-time customer?”

Target word: reverse: find the missing element **Decides:** There is no time frame.

A	Adjacent category	It names three channels and one metric.
B	Confusion set	Cost per first-time customer is measurable.
C	Absolute language	It lacks a time frame.
D	Correct	No period or deadline is given.

3. Answer A Concept 2

A grocery chain ~~with 120 stores~~ collected loyalty card data, removed duplicate and incomplete records, and fixed date formats. ~~The data is stored in a cloud warehouse.~~ What phase comes next?

Target word: next phase **Decides:** Cleaning (Process) is done.

A	Correct	After Process comes Analyze.
B	Adjacent category	The final phase, after sharing.
C	Dropped element	Skips analysis; there is nothing to share yet.
D	Inverted operation	Prepare comes before Process.

4. Answer D Concept 3

A hotel group ~~with 18 properties and a new loyalty program~~ gathers four fields per stay: guest comments on what they liked, number of nights, room type (standard or suite), and check-in wait in minutes. Which fields are qualitative?

Target word: classify data **Decides:** Descriptions and categories are qualitative.

A	Absolute language	The source does not set the data type.
B	Dropped element	Leaves out room type, a category.
C	Inverted operation	Those are quantitative.
D	Correct	Text and categories describe rather than measure.

5. Answer C Concept 3 • transfer

An analyst wants to understand why customers cancel their subscriptions. Which data best answers “why”?

Target word: reverse: pick data type for the question **Decides:** Reasons come from qualitative data.

A	Confusion set	Quantitative data show how many, not why.
B	Absolute language	Qualitative data can be coded and analyzed.
C	Correct	They capture reasons in customers’ own words.
D	Adjacent category	Show what happened, not the reasons.

6. Answer B Concept 4

A survey to a regional bank’s 9,000 mobile app users, ~~sent the week the app won a design award,~~ asks: “How much do you love our new app?” What is wrong with the question?

Target word: question flaw **Decides:** “Love” assumes a positive answer.

A	Camouflage	Timing does not fix biased wording.
B	Correct	Leading wording pushes respondents toward a favorable answer.
C	Wrong stem number	The size is not the problem.
D	Adjacent category	The core flaw is bias, not the format.

7. Answer B Concept 5

A food bank ~~with 400 volunteers~~ set the goal “reduce food waste by 15% this year.” ~~Its director drives a delivery route every Friday.~~ Which is the best KPI?

Target word: KPI for the goal **Decides:** The goal is about waste.

A	Adjacent category	A real metric, but unrelated to waste.
B	Correct	Directly measures progress on waste.
C	Confusion set	Measures output, not waste.
D	Camouflage	Irrelevant to the goal.

8. Answer B Concept 6

An insurer ~~with 1.2 million policyholders~~ keeps policy records in a relational database and receives thousands of claim photos and adjuster voice notes each week. ~~Its claims unit moved to a new office in April.~~ Which data are unstructured?

Target word: classify **Decides:** Images and audio have no row-and-column schema.

A	Absolute language	Digital does not mean structured.
B	Correct	Media files are unstructured.
C	Dropped element	Photos are unstructured too.
D	Inverted operation	Relational tables are structured.

9. Answer C Concept 6 · transfer

A dataset has a fixed schema of rows and columns and can be queried with SQL. Which term fits?

Target word: reverse: description to term **Decides:** Fixed schema in tables.

A	Confusion set	Describes content type, not structure.
B	Inverted operation	Unstructured data lack a fixed schema.
C	Correct	Rows, columns, and a schema define structured data.
D	Adjacent category	Tagged formats like JSON have flexible, not fixed, schemas.

10. Answer A Concept 7

A fulfillment center ~~that runs three shifts and ships 25,000 orders a day~~ sees late shipments rise. The team asks why: pickers are slow because items are in the wrong bins, because putaway staff skip scanning, because scanners die mid-shift, because no one is assigned to charge them. ~~The site manager holds a lean certification.~~ What root cause should be fixed?

Target word: root cause **Decides:** The last fixable why in the chain.

A	Correct	Fixing it stops the whole chain.
B	Adjacent category	A symptom near the top of the chain.
C	Confusion set	An effect of skipped scans, not the cause.
D	Dropped element	Stops a step short; they skip scans because scanners are dead.

11. Answer C Concept 8

A real estate analyst ~~at a firm founded in 1995~~ lists five sale prices on one street: \$210,000, \$225,000, \$230,000, \$240,000, and \$1.2 million for a lakefront estate. ~~The street has mature oak trees.~~ Which measure best represents a typical price, and what is it?

Target word: typical value **Decides:** One extreme outlier.

A	Confusion set	\$2,105,000 / 5; the estate pulls it far above typical.
B	Adjacent category	$(210,000 + 1,200,000) / 2$ depends only on the extremes.
C	Correct	The middle value ignores the outlier.
D	Wrong stem number	The outlier itself.

12. Answer D Concept 8 · transfer

Five salaries have a median of \$60,000. The highest salary rises from \$90,000 to \$300,000. What happens to the median and the mean?

Target word: effect of an outlier **Decides:** Median depends on the middle value; mean on the sum.

A	Inverted operation	Reverses which measure is sensitive to outliers.
B	Wrong stem number	The raise must be divided by the five salaries.
C	Dropped element	The median is not affected by the top value.
D	Correct	$\$210,000 / 5 = \$42,000$; the middle value is unchanged.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	SMART questions	A good analysis question is specific, measurable, action-oriented, relevant, and time-bound.	Accepting a vague or merely measurable question.	A stakeholder's broad wish to be reframed.
2	Phases of data analysis	Ask, Prepare, Process, Analyze, Share, Act, in that order.	Jumping from cleaning straight to sharing.	A list of steps already finished.
3	Qualitative vs. quantitative data	Quantitative data are counts and measures; qualitative data are descriptions and categories that explain why and how.	Thinking categories like room type are quantitative.	"Why" questions or text and category fields.
4	Unbiased survey questions	Questions must be neutral and not suggest an answer.	Calling a leading question merely "closed-ended."	Praise words or assumptions in the question.
5	Choosing a KPI	A KPI must measure progress toward the specific goal.	Picking a busy-looking metric unrelated to the goal.	A stated goal followed by several metrics.
6	Structured vs. unstructured data	Structured data fit rows and columns with a fixed schema; images, audio, and free text are unstructured.	Thinking anything digital is structured.	Photos, voice notes, or a relational database.
7	Root cause (5 Whys)	Keep asking why until you reach a cause you can fix that removes the problem.	Stopping at a symptom one or two steps in.	A chain of because-statements.
8	Mean vs. median	With outliers or skew, the median is the better typical value; outliers move the mean, not the median.	Reporting the mean of skewed data as typical.	One value far from the rest.

Course scope (public description): Discovering Data introduces analytical concepts, processes, and tools used in the field of business analytics. This course explores the importance of asking effective questions to collect the right data. Students will examine using analytical thinking to organize, analyze, and share data to drive decision-making. This course provides students the opportunity to work toward the Google Data Analytics Professional Certificate and the WGU Business Analysis Certificate. Introduction to Spreadsheets is a prerequisite for this course.